

Enhanced multi-leak detection in pressurized pipelines using super-resolution matched-field processing

Can Xu ^{a,b}, Muhammad Waqar ^a, Moez Louati ^c, Mohamed Salah Ghidaoui ^a

^a Department of Civil and Environmental Engineering, Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong, China

^b Ningbo Urban Construction Investment Group Co., Ltd., Ningbo, China

^c Water Supply, Sanitation and Environmental Engineering, IHE Delft Institute for Water Education, The Netherlands

ARTICLE INFO

Keywords:

Smart urban water systems
Efficient water management
Pipeline multi-leak detection
Transient waves
Matched-field processing

ABSTRACT

The detection of a cluster of leaks in pressurized water supply pipelines is crucial as it allows for early recognition of the pipeline's poor overall condition, enabling timely interventions. This paper addresses this challenge by extending, for the first time, a high-resolution modified broadband matched-field processing (MB-MFP) framework to the multi-leak detection problem. The developed methods combine advanced signal processing with acoustic measurements to achieve super-resolution identification, where leaks spaced closer than half the minimum probing wavelength are successfully resolved. A key contribution is a methodology that enables the simultaneous estimation of the number, locations, and sizes of multiple leaks. We also show that the techniques are robust to various noise types and levels, including white noise with a signal-to-noise ratio (SNR) as low as -10 dB, as well as colored and impulse noise. The effectiveness and robustness of the proposed techniques are demonstrated through comprehensive numerical simulations and validated with laboratory experiments.

1. Introduction

Identification and localization of leaks in urban water supply systems (UWSS) can prevent water loss, reduce repair costs, and mitigate environmental impacts. The U.S. Environmental Protection Agency (EPA) conducted a comprehensive assessment of existing leak detection technologies and concluded “None of the technologies was able to discern two separate leaks in close proximity ... and appear only to report the larger leak rate for the cluster of leaks. The ability to identify whether a signal is from an isolated leak or multiple leaks in close proximity is helpful in judging the general condition of a pipeline” (Nestleroth et al., 2012). In addition, a pipeline with clusters of defects (e.g., leaks) has typically lower load-bearing capacity than the same pipeline with individual defects (Zhang et al., 2023). Moreover, Lo et al. (2021) notes that the interacting defects are more detrimental towards the structural integrity of a pipeline than single defects. OMS Ltd (2025) also stated that if clustered defects appear and are left untreated, they can progressively spread over time, link together, and lead to a severe pipe failure. Therefore, there is a critical need to develop methods that can identify defects that are in close proximity.

In recent years, numerous technologies exist to detect leaks, examples include model-based (Waqar et al., 2023; Wu et al., 2025) and

data-based methods (Cheng and Li, 2023; Liu et al., 2024). Among of them, transient wave-based defect detection methods (TBDDMs) have gained significant attention. These methods utilize wave physics and inverse mathematics, where an intentionally generated transient wave propagates through the pipeline and reflects off defects (Guibert et al., 2023; Meniconi et al., 2024; Brunone, 1999; Meniconi et al., 2021). By recording and analyzing the transient pressure history within the system, it is possible to localize defects in the pipe system (Brunone et al., 2022; Che et al., 2021). TBDDMs have proven effective in identifying and characterizing various types of pipeline defects, including leaks, blockages, and structural weaknesses (Brunone et al., 2008; Wang et al., 2019b; Louati et al., 2020; Waqar et al., 2021; Ferrante et al., 2007, 2002; Wang and Ghidaoui, 2018a, 2019; Wang et al., 2019a; Wang, 2021; Lin et al., 2021; Che et al., 2022a,b).

The authors have proposed transient-based modified broadband matched-field processing (MB-MFP) techniques for single-leak detection (Xu et al., 2024). We demonstrated how the spatial positioning of side-lobes in the defect detection functional (DDF) of matched-field processing (MFP) varies with candidate leak location and probing wave frequency. As frequency increases, the side-lobe paths converge toward the actual leak location. Utilizing this property, we proposed a suite

* Corresponding author at: Department of Civil and Environmental Engineering, Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong, China.

E-mail address: cxuay@connect.ust.hk (C. Xu).

<https://doi.org/10.1016/j.watres.2025.124855>

Received 22 January 2025; Received in revised form 23 September 2025; Accepted 23 October 2025

Available online 1 November 2025

0043-1354/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature	
A	Cross-sectional area of the pipe
a	Wave speed
B	Defect detection functional of MFP
$B_{MB}, B_{MB}^1, B_{MB}^2$	Output function of MB-MFP, MB-MFP ¹ , and MB-MFP'
D	Pipe internal diameter
f	Frequency
f_D	Darcy-Weisbach friction factor
f_{th}	Fundamental frequency
G	Green's function
g	Gravitational acceleration constant
h	Pressure head perturbation
Δh	Measured leak-induced head difference vector
J	Total number of used frequencies
H_1, H_2	Pressure heads at the reservoirs
K	Total number of the assumed leaks in the model
L	Effective pipe length
N	Total number of measurement locations (i.e., sensors)
$\mathbf{n}$	Noise vector
P	Total number of the real leaks
Q_0	Steady-state discharge of the main pipe
Q_0^L, H_0^L	Steady-state discharge and pressure head at the leak
q	Discharge perturbation
$s^{L*}, \bar{s}^L$	Real and predicted leak size
$x^L, x^{L*}, \bar{x}^L$	Candidate, real, and predicted leak location, respectively
x^M	Measurement location
λ_{min}	Minimum probing wavelength
ω	Angular frequency
ω_{th}	Fundamental frequency
DDF	Defect detection functional
MB-MFP	Modified broadband matched-field processing
SNR	Signal-to-noise ratio
UWSS	Urban water supply systems
TBDDMs	Transient wave-based defect detection methods

of MB-MFP strategies that achieved high-resolution leak localization in low SNR environments with various types of noise while suppressing undesirable side-lobes.

In this paper, we seek to extend these methods to multi-leak detection, for the cluster of leaks where the leaks are closely spaced. In the context of wave propagation, “closely spaced” refers to distances between consecutive leaks that are smaller than the diffraction limit, i.e., half the wavelength of the probing wave (Fink, 2006; Waqar et al., 2022). While our previous work established the foundation for high-resolution single-leak localization, extending this framework to multi-leak scenarios presents several new and non-trivial challenges. First, the measured pressure signal becomes a complex superposition of responses from multiple leaks, and disentangling these overlapping signals to identify individual sources is a significant deconvolution problem. Second, the search for K candidate leaks must be performed in a high-dimensional parameter space, which dramatically increases the computational complexity compared to the one-dimensional search for a single leak. Third, The presence of multiple leaks inherently lowers the signal-to-noise ratio for each individual leak's contribution. Distinguishing the complex, superimposed leak signal from sophisticated environmental noise (e.g., white or non-white noise) requires a significantly more robust methodology. Finally, and most critically, the true number of leaks (P) is generally unknown in

practice, necessitating a methodology that can estimate this number rather than assuming it a priori. This paper builds upon the foundational MB-MFP technique to specifically address these challenges, providing a comprehensive solution for multi-leak detection.

The structure of this paper is as follows. Section 2 introduces the frequency-domain transient wave model for pipe systems and explains how the head difference signal, generated by leaks during bidirectional wave propagation, is extracted for detection. Section 3 details the proposed MB-MFP techniques for multi-leak localization. The performance of these methods is then evaluated through comprehensive numerical simulations, demonstrating its accuracy and super-resolution capabilities under various noise conditions and for closely spaced leaks with different leak numbers. Section 4 investigates the sensitivity of the proposed methods when the assumed number of leaks in the model (K) differs from the true number of leaks (P), showing that for $K \geq P$, the estimated locations reliably converge to the actual leak positions. Section 5 experimentally validates the MB-MFP techniques in a laboratory setting featuring dual leaks of varying sizes. Finally, Section 6 summarizes the key findings and provides concluding remarks.

2. Model description

Consider a pipeline system with a cluster of P leaks. Let q and h respectively denote the transient-induced fluid discharge oscillation and pressure head oscillation. The measured head response at the measurement point x^M and the frequency ω to an actively generated wave is given as follows:

$$h^M(\omega, x^M, \mathbf{x}^L) = h^{NL}(\omega, x^M) + \underbrace{\sum_{p=1}^P s_p^{L*} G(\omega, x^M, x_p^{L*})}_{\text{the scatterings from each leak}} + n(\omega, x^M) \quad (1)$$

which is obtained by solving the frequency-domain mass and momentum conservation equations (Wang and Ghidaoui, 2018b). h^M is constituted of three distinct components: (1) the baseline head h^{NL} where superscript “NL” denotes “no leak”, (2) the scattering arising from p th leak and expressed by multiplying the Green's function G regarding to the leak location x_p^{L*} with the leak size s_p^{L*} , where the superscript L^* represents the real leaks, and (3) the noise n . In fact, there originally existed a fourth component in Eq. (1), which involves the pressure heads resulting from the coupling interaction between multiple leaks. These coupling effects are neglected since they are relatively insignificant compared to the scattering originating from each leak, under the assumption of small leaks (i.e., $s_p^{L*} \ll 1 \text{ m}^2, p = 1, \dots, P$) which are a subject of interest for transient-based methods (Wang and Ghidaoui, 2018a). For example, in a two-leak case (as depicted in Fig. 1), the transient wave propagation includes three routes: the baseline head h^{NL} , the scatterings $s_1^{L*} G(x_1^{L*})$ and $s_2^{L*} G(x_2^{L*})$, and the negligible coupling effect. In Eq. (1), h^{NL} is expressed as

$$h^{NL}(\omega, x^M) = -Z \sinh(\mu x^M) q^U(\omega) + \cosh(\mu x^M) h^U(\omega) \quad (2)$$

and G is given by

$$G(\omega, x^M, x_p^{L*}) = -\frac{\sqrt{g} Z \sinh(\mu(x^M - x_p^{L*}))}{\sqrt{2H_0^L}} \begin{pmatrix} Z \sinh(\mu x_p^{L*}) q^U(\omega) \\ -\cosh(\mu x_p^{L*}) h^U(\omega) \end{pmatrix} \quad (3)$$

Eqs. (2) and (3) indicate the relation between h^M at the measurement point x^M and (q^U, h^U) at the upstream end x^U ; $Z = (\mu a^2)/(i\omega g A)$ is the characteristic impedance of pipe, which is determined by the wave speed a , frequency ω , gravitational acceleration g , cross-sectional area A , and propagation function μ defined as $\mu = a^{-1}(-\omega^2 + i g A \omega R)^{1/2}$; R is the linearized steady-state resistance term and $R = (f_D Q_0)/(g D A^2)$; f_D represents the Darcy–Weisbach friction factor and Q_0 denotes the

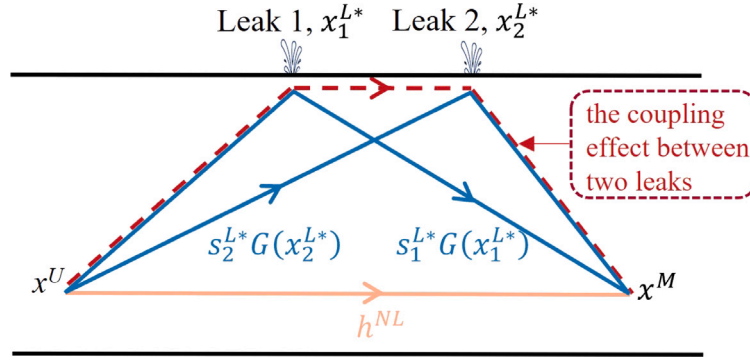


Fig. 1. Transient wave propagation in a two-leak case.

steady-state discharge at the pipe upstream end; D is the pipeline internal diameter. For pipeline made of high-density polyethylene (HDPE), the effective wave speed can be formulated as

$$a(\omega; \boldsymbol{\theta}) = \left[\frac{1}{a_0^2} + \frac{\rho D}{e} (1 - \nu^2) \sum_{k=1}^{N_{kv}} \frac{J_k}{1 + i\omega\tau_k} \right]^{-1/2} \quad (4)$$

which depends on the frequency ω and incorporates the viscoelastic effects by using the Kelvin–Voigt (K-V) model, where J_k and τ_k are respectively the creep compliance coefficient and retardation time coefficient in the k th K-V model element; N_{kv} is the total number of K-V model elements; a_0 represents the elastic component of wave speed and defined as

$$a_0 = \left(\frac{\rho}{\kappa} + (1 - \nu^2) \frac{\rho D}{eE} \right)^{-1/2} \quad (5)$$

where $\rho \equiv$ Water density, $\kappa \equiv$ Bulk modulus of water, $e \equiv$ Pipe wall thickness, $\nu \equiv$ Poisson's ratio, $E \equiv$ Young's modulus of elasticity of pipe wall.

By taking the difference between h^M and h^{NL} in Eq. (1), the leak-induced pressure head Δh at ω and x^M can be defined and expressed as

$$\begin{aligned} \Delta h(\omega, x^M, \mathbf{x}^{L*}) &= h^M(\omega, x^M, \mathbf{x}^{L*}) - h^{NL}(\omega, x^M) \\ &= \sum_{p=1}^P s_p^{L*} G(\omega, x^M, x_p^{L*}) + n(\omega, x^M) \end{aligned} \quad (6)$$

which is a linear combination of individual leak contributions. Then, Δh across all J frequencies and N measurement locations can be written as a vector form:

$$\begin{aligned} \Delta \mathbf{h}(\omega, \mathbf{x}^M, \mathbf{x}^{L*}) &= \mathbf{h}^M(\omega, \mathbf{x}^M, \mathbf{x}^{L*}) - \mathbf{h}^{NL}(\omega, \mathbf{x}^M) \\ &= \sum_{p=1}^P s_p^{L*} \mathbf{G}(\omega, \mathbf{x}^M, x_p^{L*}) + \mathbf{n}(\omega, \mathbf{x}^M) \end{aligned} \quad (7)$$

wherein $\Delta \mathbf{h}$, $\mathbf{h}^M$, $\mathbf{h}^{NL}$, $\mathbf{G}$, and $\mathbf{n}$ are all $N \cdot J$ dimensional complex vectors and their forms can be illustrated by considering an example:

$$\begin{aligned} \mathbf{G}(\omega, \mathbf{x}^M, x_p^{L*}) &= \left[G(\omega_1, x_1^M, x_p^{L*}), \dots, G(\omega_1, x_N^M, x_p^{L*}), \dots, \right. \\ &\quad \left. G(\omega_J, x_1^M, x_p^{L*}), \dots, G(\omega_J, x_N^M, x_p^{L*}) \right]^{\dagger} \end{aligned} \quad (8)$$

where the superscript $\dagger$ denotes vector transpose. The purpose of obtaining the measured head difference vector $\Delta \mathbf{h}$ is to facilitate leak localization. Note that the first equal sign in Eq. (7) signifies that $\Delta \mathbf{h}$ can be determined from the pressure measurement $\mathbf{h}^M$, and the second equal sign represents the theoretical model of $\Delta \mathbf{h}$, incorporating leak parameters s^{L*} and $\mathbf{x}^{L*}$.

This paper examines a reservoir-pipe-valve (RPV) system featuring multiple leaks, as illustrated in Fig. 2. The system consists of a single pipe, positioned horizontally and extending a length L , which connects two reservoirs at each end. These reservoirs are characterized by pressure heads H_1 and H_2 . A valve is positioned near the downstream end and can be used to generate transient waves by rapid “close-open” maneuvers. An example of such a maneuver is shown in Fig. 3, where a discharge perturbation with a pulse shape is introduced. The resulting transient wave propagates back and forth within the pipe at a wave speed of a . The system's fundamental period and frequency are respectively given by $T_{th} = 4L/a_0$ and $f_{th} = 1/T_{th}$ (or $\omega_{th} = 2\pi/T_{th}$) (Chaudhry, 2014). The utilized angular frequency range is denoted as $\boldsymbol{\omega} = [\omega_1, \dots, \omega_J]^{\dagger}$ and the corresponding frequency range is $\mathbf{f} = [f_1, \dots, f_J]^{\dagger}$, where J represents the number of excited frequencies. Our study considers P leaks with sizes $s^{L*} = [s_1^{L*}, \dots, s_P^{L*}]^{\dagger}$ and locations $\mathbf{x}^{L*} = [x_1^{L*}, \dots, x_P^{L*}]^{\dagger}$. The steady-state discharge through the p th ($p \in [1, \dots, P]$) leak is determined using the orifice equation: $Q_{0,p}^L = s_p^L \sqrt{2gH_{0,p}^L}$, where $H_{0,p}^L$ represents the steady-state head at the leak location, calculated as $H_{0,p}^L = H_1 - (H_1 - H_2)(x_p^L/L)$. The number and distribution of measurement locations are denoted as N and $\mathbf{x}^M = [x_1^M, \dots, x_N^M]^{\dagger}$, respectively. Previous experience (Xu et al., 2024) showed that $N = 2$ is sufficient to provide adequate information, although additional measurement locations can yield more valuable insights. The distance between two measurement locations is intentionally chosen to be larger than half of the minimum probing wavelength (i.e., $0.5\lambda_{min} = 0.5a_0/f_J$), which represents the spatial resolution limit of the propagating waves due to wave diffraction effect (Fink, 2006; Waqar et al., 2022).

To incorporate noise into numerical simulations, the signal-to-noise ratio (SNR) in decibels is defined as

$$\text{SNR} = 20 \log_{10} \left(\frac{\mathbb{E} \left(\left| \sum_{p=1}^P s_p^{L*} \mathbf{G}(\omega, \mathbf{x}^M, x_p^{L*}) \right|^2 \right)}{\sigma^2} \right) \quad (9)$$

where $\mathbb{E}[\cdot]$ denotes the mean head difference, and σ represents the standard deviation of Gaussian white noise with zero mean. In pipeline systems, noise can be categorized as white noise, colored noise, or impulse-type noise (Dubey et al., 2019). For white noise, SNR simply remains a constant across the frequency range $\boldsymbol{\omega}$. For blue noise, SNR decreases by 3 dB per increasing frequency from SNR_1 at ω_1 (the starting frequency). Regarding impulse noise, SNR is constant (e.g., $\text{SNR} = 10$ dB) over $\boldsymbol{\omega}$, except for specific frequencies (e.g., $\omega_j = z \cdot 0.1\omega_{th}$, $z \in \mathbb{Z}^+$), at which the SNR abruptly drops to a much lower value, denoted as SNR_2 .

3. Multi-leak detection by MB-MFP techniques

This section extends the scope of using MB-MFP techniques beyond single-leak detection to address the challenge of detecting multiple

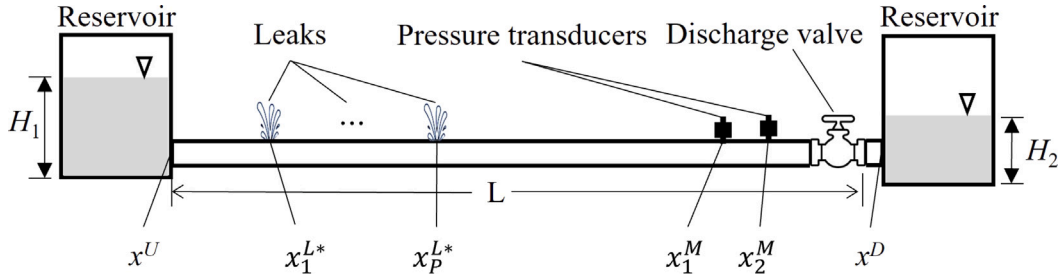


Fig. 2. Schematic of the RPV model with multiple leaks and measurement locations.

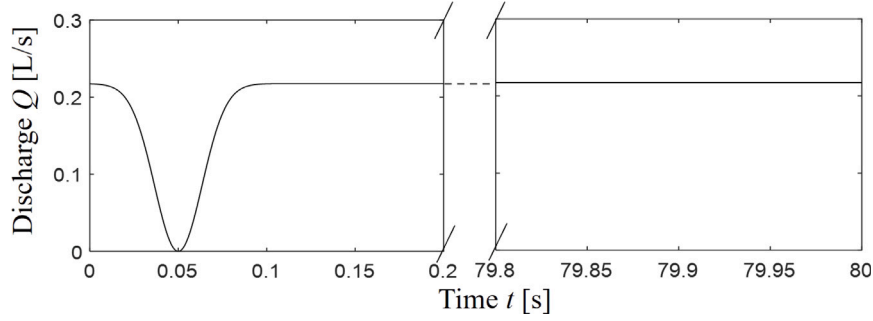


Fig. 3. Downstream-end discharge undergoes pulse-shaped changes due to valve operations.

leaks. This extension aims to achieve super-resolution localization, even in the 2-leak detection scenario where the white noise with a signal-to-noise ratio (SNR) as low as -10 dB and the leaks are in close proximity (i.e., distances less than $0.5\lambda_{\min}$).

3.1. MB-MFP techniques

The concept of MFP with respect to frequency and candidate leak location is first introduced, with a focus on its definition and the underlying mathematical framework. The goal is to provide a better understanding of how the subsequent MB-MFP techniques successfully localize leaks. Similar to the definition given in Xu et al. (2024), the defect detection functional (DDF) of MFP is defined as:

$$|B(\omega_j, \mathbf{x}^L)| = \frac{|\langle \Delta \mathbf{h}_j, \mathbf{G}_j \rangle|}{\|\Delta \mathbf{h}_j\| \|\mathbf{G}_j\|} = \frac{|\Delta \mathbf{h}_j^H \mathbf{G}_j|}{\sqrt{\Delta \mathbf{h}_j^H \Delta \mathbf{h}_j} \sqrt{\mathbf{G}_j^H \mathbf{G}_j}} \quad (10)$$

where the measured data vector $\Delta \mathbf{h}_j$ and the modeled data vector $\mathbf{G}_j$ are both N -dimensional and include information from N given measurement locations at j th frequency ω_j ($j \in [1, \dots, J]$). In this paper where multiple leaks are studied, the expressions of $\Delta \mathbf{h}_j$ and $\mathbf{G}_j$ can be obtained from Eqs. (5) and (6) as:

$$\Delta \mathbf{h}_j = \Delta \mathbf{h}(\omega_j, \mathbf{x}^M, \mathbf{x}^{L*}) = \sum_{p=1}^P s_p^{L*} \mathbf{G}(\omega_j, \mathbf{x}^M, x_p^{L*}) + \mathbf{n}_j \quad (11)$$

and

$$\mathbf{G}_j = \sum_{k=1}^K \mathbf{G}(\omega_j, \mathbf{x}^M, x_k^L) \quad (12)$$

where K is the assumed number of candidate leaks in the model and P is the real number of leaks. That is, in Eq. (10), a K -leak model is used to detect the P -leak case. To simplify the problem, this section considers only the case of $K = P$. The situation where $K \neq P$ is discussed in the next section.

When $K = P$ and noise is not considered, $|B|$ reaches the maximum value of one at positions $\mathbf{x}^L = \mathbf{x}^{L*}$ at each single frequency (Xu et al., 2024). Consistently, after integrating $|B|$ over all frequencies, the resulting modified broadband MFP (MB-MFP) method exhibits maxima

at $\mathbf{x}^L = \mathbf{x}^{L*}$. The functional expression for MB-MFP based on a K -leak model, i.e., MB-MFP(K) [$|B_{MB}(K)|$] is provided in integral form as:

$$|B_{MB}(\mathbf{x}^L)| = \frac{1}{\omega_{\max} - \omega_{\min}} \left| \int_{\omega_{\min}}^{\omega_{\max}} B(\omega_j, \mathbf{x}^L) d\omega \right| \quad (13)$$

To suppress the appeared and undesired side-lobes shown in the MB-MFP result and thus enhance the uniqueness of leak localization results, we introduce an optimized broadband MFP method—MB-MFP'—leveraging gradient information from the output surface of $|B|$ on the defect location-frequency plane. Stated briefly, MB-MFP' uses a peak-extraction algorithm, assigning zero values to points adjacent to the maxima of $|B|$ (Xu et al., 2024). Then, Integrating the processed $|B|$ over frequencies allows MB-MFP' to achieve enhanced leak localization with minimized side-lobes. The functional expression of MB-MFP'(K) [$|B'_{MB}(K)|$] is given as:

$$|B'_{MB}(\mathbf{x}^L)| = \frac{\int_{\omega_{\min}}^{\omega_{\max}} \left(\sum_{k=1}^K \left(\begin{array}{c} \text{sign} \left(\frac{\partial |B(\omega_j, \mathbf{x}_k^L)|}{\partial x_k^L} \right) \\ - \text{sign} \left(\frac{\partial |B(\omega_j, \mathbf{x}_k^L + \Delta \mathbf{x}_k^L)|}{\partial x_k^L} \right) \end{array} \right) \right) \cdot |B(\omega_j, \mathbf{x}^L)| d\omega}{\omega_{\max} - \omega_{\min}} \quad (14)$$

The robustness of MB-MFP and MB-MFP' is augmented by the normalization of $\Delta \mathbf{h}_j$ in $|B|$ [Eq. (10)], effectively compressing the impact of high amplitude noise $\mathbf{n}_j$. This makes the proposed broadband MFP methods effective against various noise types, including white, colored (e.g., blue noise), and impulse noise.

3.2. Numerical simulation results assuming $K = P$

In this subsection, we evaluate the anti-noise capability and side-lobe suppression performance of the MB-MFP techniques for multi-leak detection through comprehensive numerical simulations under the assumption that $K = P$ (the case of $K \neq P$ will be discussed in the next section). The simulations encompass various noise types and levels, as well as different leak intervals. The parameters of the RPV system are

Table 1

The system parameters used in the simulations.

Pipe length $L = 200$ m	Pipe diameter $D = 0.08$ m	Elastic wave speed $a_0 = 1000$ m/s
Friction factor $f_D = 0.02$	Upstream head $H_1 = 25$ m	Downstream head $H_2 = 20$ m
Simulated leak locations $\mathbf{x}^{L*} = [x_1^{L*}, \dots, x_P^{L*}]^T$ with leak size $s^{L*} = 1.4 \times 10^{-5}$ m ² (per leak)		
Measurement locations $\mathbf{x}^M = [x_1^M, x_2^M]^T = [180 \text{ m}, 200 \text{ m}]$		
Frequency range $\omega = [\omega_{th}, \omega_{th} + 0.02\omega_{th}, \dots, 20\omega_{th}]^T$, or $\mathbf{f} = [f_{th}, f_{th} + 0.02f_{th}, \dots, 20f_{th}]^T$		

summarized in Table 1. The pipe-wall is assumed to be elastic. The system's fundamental period and frequency are calculated as $T_{th} = 4L/a_0 = 0.8$ s and $f_{th} = 1/T_{th} = 1.25$ Hz (or $\omega_{th} = 2\pi/T_{th} = 7.85$ rad/s). Thus, the shortest (minimum) probing wavelength $\lambda_{min} = a_0/f_J = 40$ m. All numerical simulations were conducted using custom scripts developed in Matlab R2022b, running on a Windows 11 system with an Intel(R) Core(TM) i7-9700 CPU @ 3.00 GHz and 24.0 GB of RAM.

3.2.1. Noise-free case

Fig. 4 illustrates the results of using MB-MFP(K) and MB-MFP'(K) for the noise-free case, where $K = P = 2$. Each method successfully identifies two leaks even when the distance between them is below $0.5\lambda_{min}$, demonstrating super-resolution capabilities beyond the typical spatial resolution limit of propagating waves. Specifically, MB-MFP'(2) shows superior main-lobe resolution and side-lobe suppression compared to MB-MFP (see the comparison between Fig. 4(a-1) and (a-2), or between Fig. 4(b-1) and (b-2)). Since these simulations are noise-free, analogous results are obtained for cases with more than two leaks ($K = P > 2$). In such scenarios, both MB-MFP and MB-MFP' accurately localize all leaks, with MB-MFP' consistently demonstrating superior localization accuracy.

3.2.2. Various noise scenarios

This sub-section presents numerical examples under various noise scenarios to evaluate the robustness and scalability of the proposed methods. We consider white noise with $\text{SNR} = [-30, -20, \dots, 20]$ dB, blue noise with $\text{SNR}_1 = [0, 10, \dots, 50]$ dB, and impulse noise with $\text{SNR}_2 = [-100, -80, \dots, 0]$ dB. Each scenario is repeated twenty times to compute the average leak localization error, defined as the L2-error between estimated and real leak locations:

$$e(\bar{\mathbf{x}}^L) = \|\bar{\mathbf{x}}^L - \mathbf{x}^{L*}\|_2 = \sqrt{(\bar{x}_1^L - x_1^{L*})^2 + \dots + (\bar{x}_K^L - x_K^{L*})^2} \quad (15)$$

where $\bar{\mathbf{x}}^L = [\bar{x}_1^L, \dots, \bar{x}_K^L]^T$ is the vector of estimated leak locations and we have assumed $K = P$ in this subsection. Fig. 5 shows the localization errors of using MB-MFP(K) and MB-MFP'(K) for $K = P = 1, 2, 3$, and 4. Several key observations can be made:

(1) For a given signal-to-noise ratio (SNR), the leak localization error increases as the number of leaks increases. For example, as shown in the new Fig. 5(a) for white noise at an SNR of -10 dB, the MB-MFP method yields errors of less than 2 m for one-leak and two-leak cases, while the errors grow to approximately 4 m and 11 m for the three-leak and four-leak cases, respectively. At SNR = -20 dB, this trend becomes even more pronounced: MB-MFP(1) still achieves highly accurate localization with an error below 2 m, while MB-MFP(2), MB-MFP(3), and MB-MFP(4) completely fail to localize the leaks. A similar pattern can also be observed in the localization results obtained with MB-MFP'. These trends are expected, as resolving a greater number of leaks requires disentangling a more complex superposition of signals, making the inverse problem more sensitive to noise.

(2) In general, while MB-MFP' offers higher resolution, its noise resistance is slightly weaker than that of MB-MFP, particularly under colored (blue) noise (as shown in Fig. 5(b)). This highlights a classic trade-off between resolution and noise immunity. Nevertheless, both methods exhibit excellent overall robustness.

(3) As shown in Fig. 5(c), both methods demonstrate excellent robustness to impulse noise, primarily because the signal normalization

step in our algorithm effectively suppresses its characteristic high-amplitude spikes at specific frequencies.

Fig. 6 shows the results for a special case of two closely placed leaks, only $0.2\lambda_{min}$ apart. When compared to the two-leak detection case with a larger interval (i.e., $|x_1^{L*} - x_2^{L*}| = 50$ m = $1.25\lambda_{min}$ in Fig. 5), both MB-MFP and MB-MFP' can still successfully identify the leaks and maintain similar high accuracy.

4. A methodology to determine the unknown leak number

This section investigates a methodology to determine the unknown number of leaks, building upon the earlier discussion where the assumption $K = P$ was explored. In real-world applications, accurately knowing the number of leaks (P) in advance is often impractical, which prompts the examination of scenarios where $K \neq P$. Our analysis spans scenarios with $K < P$, $K = P$, and $K > P$. It is found that the K estimated leak locations tend to cluster around the P actual leak locations as K reaches and eventually surpasses P . This clustering property provides a systematic approach to concurrently estimating the number of leaks while precisely determining their locations.

Following Wang and Ghidaoui's methodology (Wang and Ghidaoui, 2019), the leak sizes can be calculated as

$$\bar{s}^L = (\tilde{\mathbf{G}}^H \tilde{\mathbf{G}})^{-1} \tilde{\mathbf{G}}^H \Delta \mathbf{h} \quad (16)$$

and

$$\tilde{\mathbf{G}}(\omega, \mathbf{x}^M, \bar{\mathbf{x}}^L) = [\mathbf{G}(\omega, \mathbf{x}^M, \bar{x}_1^L) \dots \mathbf{G}(\omega, \mathbf{x}^M, \bar{x}_K^L)] \quad (17)$$

once we determine the leak locations, where $\tilde{\mathbf{G}}$ is an $N \cdot J \times K$ matrix, $\bar{\mathbf{x}}^L = [\bar{x}_1^L, \dots, \bar{x}_K^L]$ and $\bar{s}^L = [\bar{s}_1^L, \dots, \bar{s}_K^L]$ are K -dimensional vectors and denotes the K estimated leak locations and sizes. Fig. 7 illustrates leak detection results for $P = 2$ using MB-MFP(K) with $K = 1, 2, 3$, and 4. When $K < P$, Fig. 7(a) shows that MB-MFP incorrectly point out leak locations. Conversely, when K matches P , as shown in Fig. 7(b), both leak locations and sizes are estimated accurately. Moreover, when K exceeds P , as depicted in Fig. 7(c) and (d), the estimated K leak locations converge near the P real leak locations and the summation of the calculated leak sizes at the same region is close to the real leak size at the region.

The clustering property suggests that modeling with $K \geq P$ gives the system enough "capacity" to handle all possible leak events, similar to the concept of "overcomplete representation" commonly used in signal processing and machine learning for enhancing stability and robustness of results (Livezey et al., 2016). An overcomplete representation, where the number of latent features exceeds the dimensions of the signal, is particularly valuable when the signal comprises multiple sets of complete linear features. In pipeline systems, where the model is linearized to account for multiple leaks, this over-completeness holds true. This property not only aids in precise leak localization, but also facilitates the estimation of the number of leaks.

In fact, the observed clustering phenomena are consistent with the results discussed in Wang and Ghidaoui's work (Wang and Ghidaoui, 2019). In their work, the number of leakages can then be determined by calculating the log-likelihood function:

$$\log L(K, \bar{\mathbf{x}}^L, \bar{s}^L | \Delta \mathbf{h}) = - \left\| \Delta \mathbf{h} - \sum_{k=1}^K \mathbf{G}(\omega, \mathbf{x}^M, \bar{x}_k^L) \bar{s}_k^L \right\|_2^2 \quad (18)$$

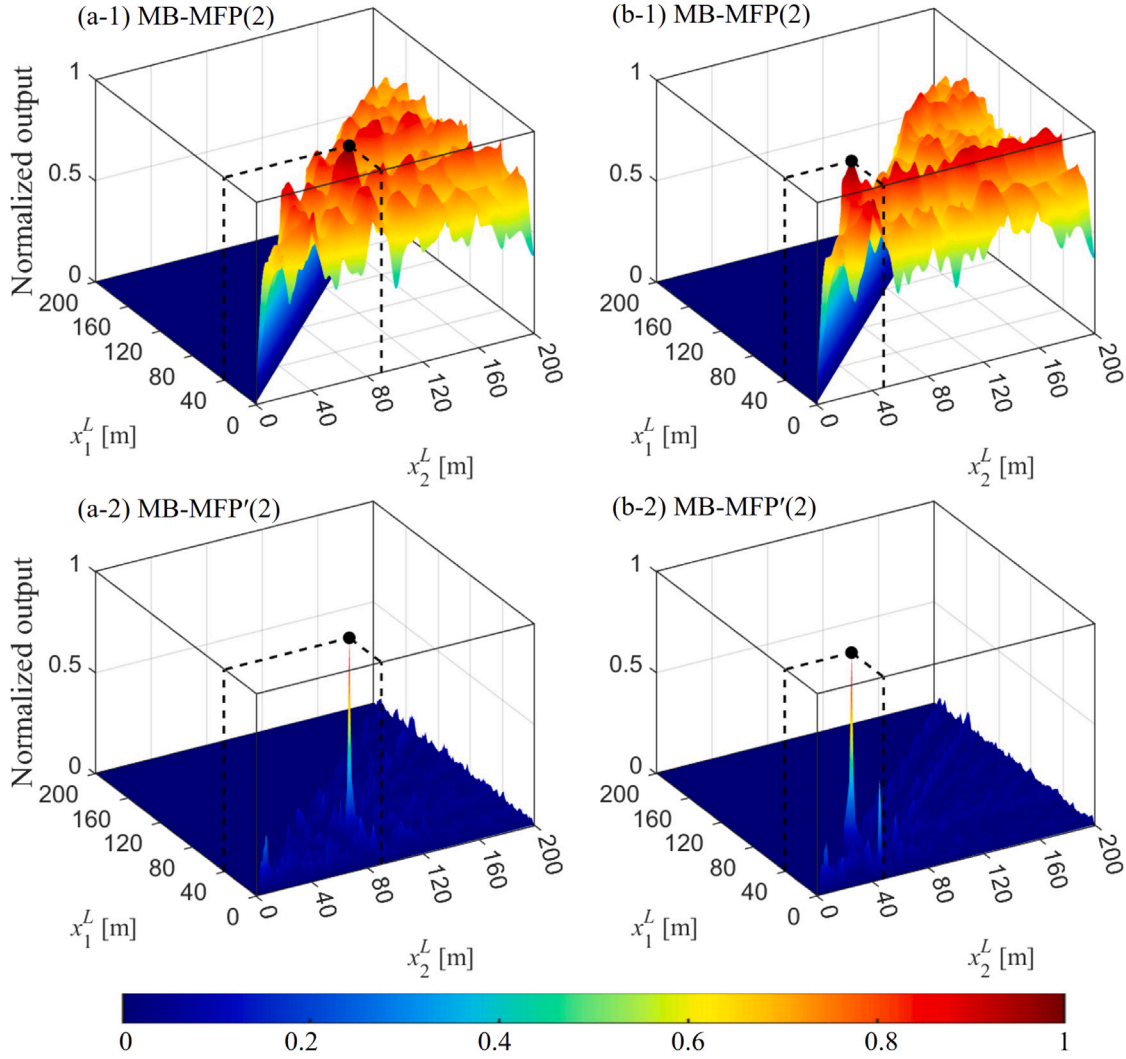


Fig. 4. Demonstrations of using MB-MFP(2) [Eq. (13)] and MB-MFP'(2) [Eq. (14)] for two-leak detection cases: (a) $\mathbf{x}^{L*} = [40\text{ m}, 90\text{ m}]$, and the space between two leaks is $|\mathbf{x}_1^{L*} - \mathbf{x}_2^{L*}| = 50\text{ m} = 1.25\lambda_{\min}$ (b) $\mathbf{x}^{L*} = [40\text{ m}, 48\text{ m}]$, and the space between two leaks is $|\mathbf{x}_1^{L*} - \mathbf{x}_2^{L*}| = 8\text{ m} = 0.2\lambda_{\min}$. In all plots, the black dot represents the real leak locations.

Fig. 8 present the calculated log-likelihood results. For the case of $P = 2$, it can be observed that the log-likelihood increases rapidly when K gets close to $P = 2$ and then varies slightly and converges when $K \geq P = 2$, indicating that the correct number of leaks is two. This align with the situation presented in Fig. 7.

5. Experimental validation in a laboratory setup

This section presents the experimental validation of the proposed MB-MFP techniques in a controlled laboratory setup at the Water Resources Research Laboratory of HKUST. It is important to note that the objective of this experiment is not to replicate the exhaustive parametric study of the numerical simulations, but rather to serve as a proof-of-principle. The goal is to demonstrate that the key phenomena and the superior performance of the proposed methods, predicted under idealized numerical conditions, hold true in a real-world physical system. This system introduces inherent complexities, such as the viscoelastic behavior of the High-Density Polyethylene (HDPE) pipe wall (accounted for by the Kelvin–Voigt model in Eq. (4)), uncharacterized measurement noise, and non-ideal transient generation, thus providing a robust test of the methods' practical applicability.

The facility comprises a pipeline system, illustrated through a comprehensive 3D schematic and a simplified 2D layout as shown in Fig.

9. This pipeline system includes essential components such as a booster pump at the upstream end and a valve at the downstream end. These key components of the experimental setup are summarized in Table 2. The experimental loop in Fig. 9(a) comprises straight pipe runs connected by long-radius bends and standard couplings. Bend angles range from 70° to 90° and radii of curvature (R_c) from 0.20 m to 0.30 m, chosen to limit form losses. In the steady state, minor losses due to fittings are accounted for by calibrating an effective Darcy–Weisbach friction factor to match the measured total head loss. For the transients considered here, the injected signals are band-limited with a minimum wavelength $\lambda_{\min} = a_0/f_J \approx 61\text{ m}$ (the choice of the maximum frequency f_J is based on the SNR of the measured signals along frequency), which is much larger than the characteristic bend radius, yielding $kR_c = (2\pi/\lambda_{\min})R_c \approx 0.03$ where k is the wavenumber. This scale separation implies negligible wave scattering by curvature at the excited frequencies. Additionally, the test rig is restrained to the floor with T-slot extrusion bars and U-shaped clamps to prevent lateral motion during transient events.

Strategically positioned along the pipeline are several manual ball valves (MVs) used either to install pressure transducers or to simulate leaks. In our tests, four MVs are chosen, i.e., MV 1 for generating transient waves by rapid-closure maneuver, and MV 2, MV 4, and MV 8 for data collection purposes. The pipe is made of high-density

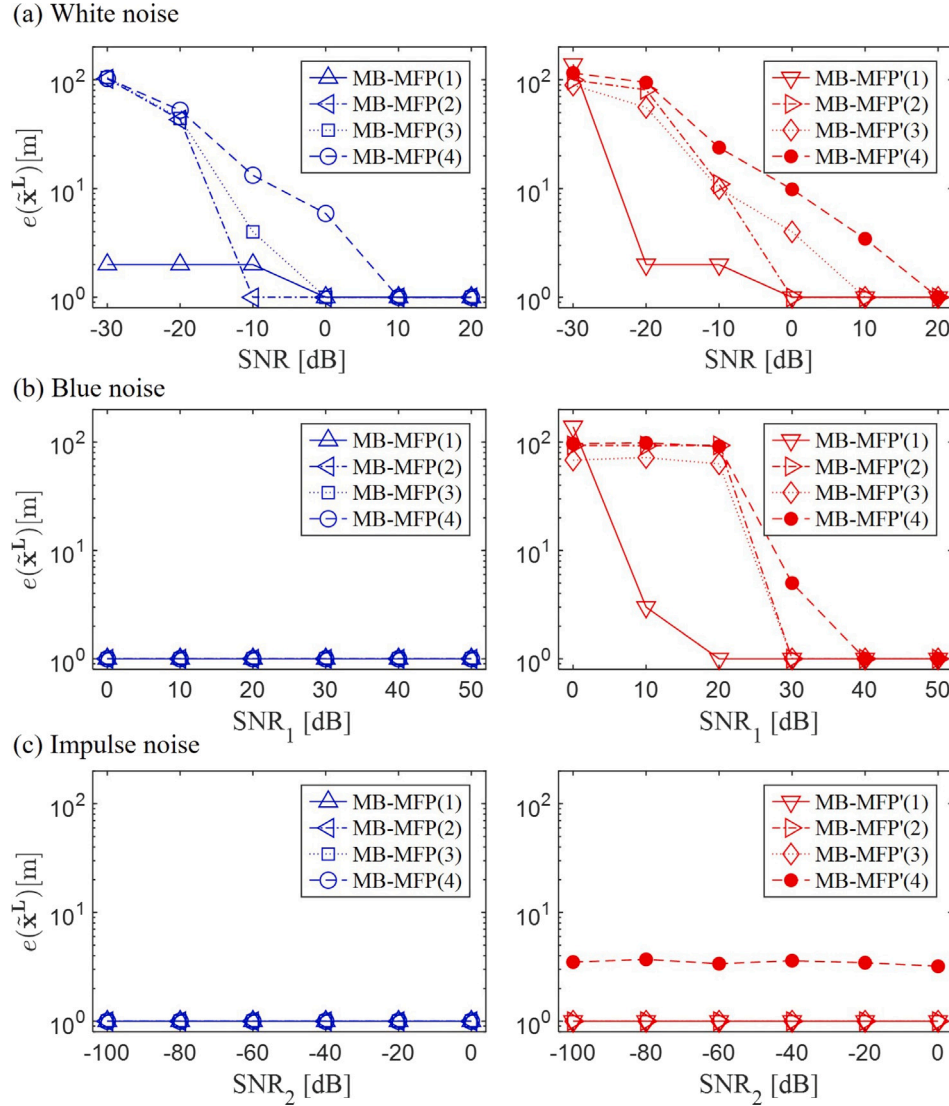


Fig. 5. Leak localization errors of using MB-MFP(K) and MB-MFP'(K) for $K = P = 1$ ($x^{L*} = 40$ m), $K = P = 2$ ($x^{L*} = [40$ m, 90 m]), $K = P = 3$ ($x^{L*} = [40$ m, 60 m, 90 m]), and $K = P = 4$ ($x^{L*} = [40$ m, 60 m, 90 m, 120 m,]) under different noise types and levels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

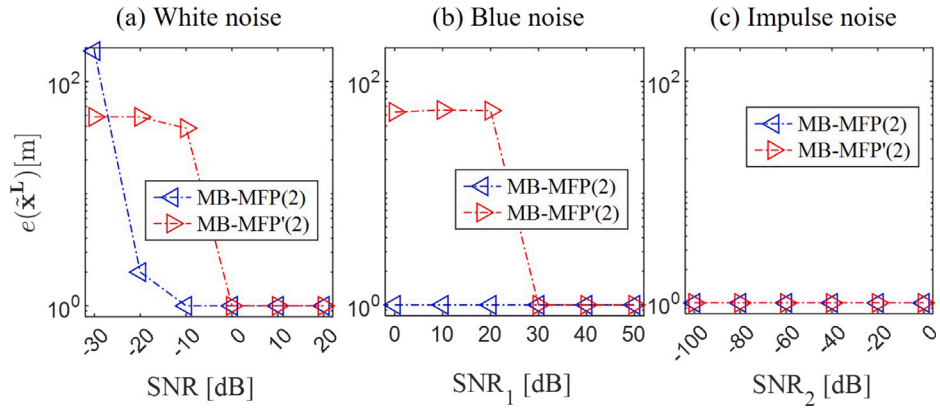


Fig. 6. Leak localization errors of using MB-MFP(2) and MB-MFP'(2) for two-leak detection cases with $x^{L*} = [x_1^{L*}, x_2^{L*}] = [40$ m, 48 m] (i.e., $|x_1^{L*} - x_2^{L*}| = 4$ m = $0.2\lambda_{\min} < 0.5\lambda_{\min}$) under different noise scenarios.

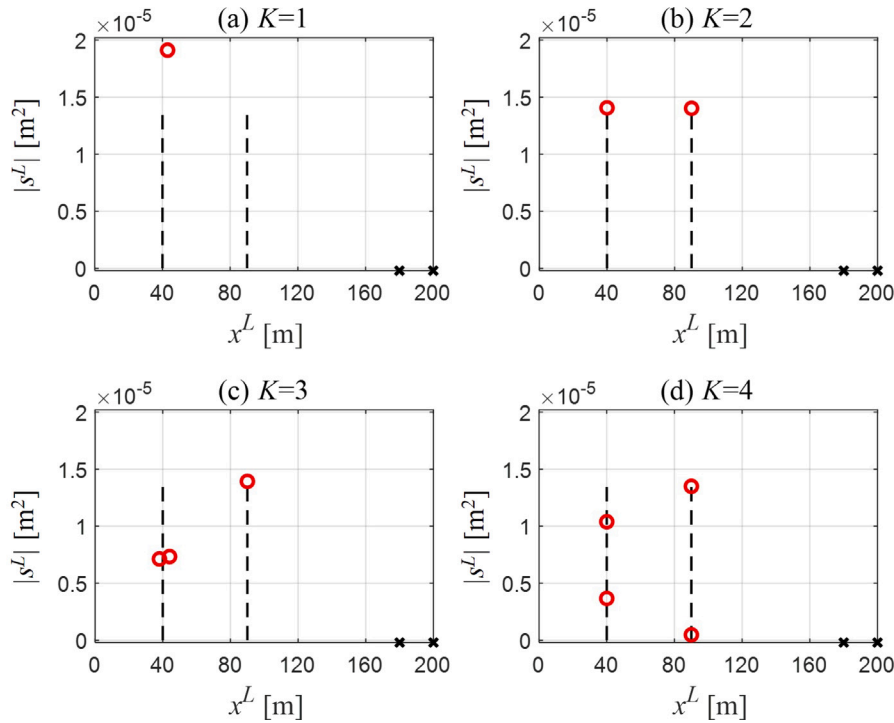


Fig. 7. Two-leak ($P = 2$) detection case by using MB-MFP with modeled leak number $K = 1, 2, 3$ and 4 . The real leak locations and sizes are respectively $\mathbf{x}^{L*} = [40, 90] \text{ m}$ and $\mathbf{s}^{L*} = [1.4, 1.4] \times 10^{-5} \text{ m}^2$, and represented by the dashed vertical lines. The red circles represent the estimated leaks. The estimated leak locations and sizes are (a) $\hat{\mathbf{x}}^L = 43 \text{ m}$, $\hat{\mathbf{s}}^L = 1.912 \times 10^{-5} \text{ m}^2$, (b) $\hat{\mathbf{x}}^L = [40, 90] \text{ m}$, $\hat{\mathbf{s}}^L = [1.406, 1.402] \times 10^{-5} \text{ m}^2$, (c) $\hat{\mathbf{x}}^L = [38, 44, 90] \text{ m}$, $\hat{\mathbf{s}}^L = [0.713, 0.734, 1.393] \times 10^{-5} \text{ m}^2$, (d) $\hat{\mathbf{x}}^L = [40, 40, 90, 90] \text{ m}$, $\hat{\mathbf{s}}^L = [0.367, 1.039, 0.048, 1.350] \times 10^{-5} \text{ m}^2$. The black crosses represent the measurement locations.

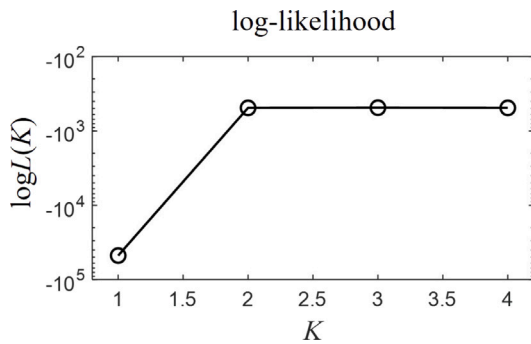


Fig. 8. The log-likelihood of $K = 1, 2, 3$ and 4 . The real leak number $P = 2$.

polyethylene, and its specific viscoelastic parameters are listed Table 3. These parameters are instrumental in calculating the effective wave speed (Eq. (4)) across various frequencies. The other system parameters are also given in Table 3 for easy reference. The steady-state flow rates through the main pipe and at the two leaks are measured, being approximately $Q_0 \approx 1.47 \times 10^{-3} \text{ m}^3/\text{s}$, $Q_0^{L1} = 0.40 \times 10^{-3} \text{ m}^3/\text{s}$, and $Q_0^{L2} = 0.63 \times 10^{-3} \text{ m}^3/\text{s}$. The system's fundamental period and frequency are calculated as $T_{th} = 4L/a_0 = 1.578 \text{ s}$ and $f_{th} = 1/T_{th} = 0.6337 \text{ Hz}$. The leaks are simulated at locations $\mathbf{x}^{L*} = [44.09 \text{ m}, 67.83 \text{ m}]$. Different leak sizes are set to validate the MB-MFP techniques' ability to simultaneously detect a smaller leak alongside a larger one.

A rapid full closure of valve MV 1 within 0.05 s initiates the transient wave generation. Subsequently, pressure signals are acquired at MV 2, MV 4, and MV 8 in the time domain and transformed into the frequency domain, as depicted in Fig. 10. Notably, these signals exhibit significant noise levels beyond the 6 Hz frequency range, leading to the selection of $f_J = 6 \text{ Hz}$ as the upper-frequency limit for subsequent analyses, corresponding to a minimum probing wavelength of $\lambda_{\min} =$

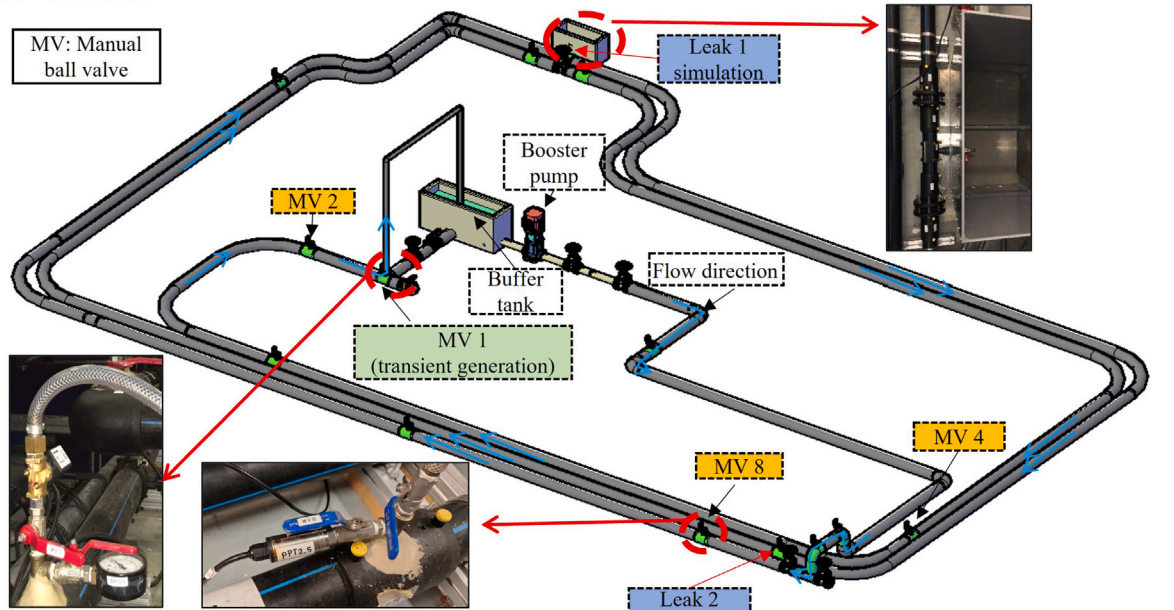
$a_0/f_J \approx 61 \text{ m}$. The signal at MV8 is employed in conjunction with the transfer matrix to compute the discharge at the upstream end. Meanwhile, the signals at MV2 and MV4 are used for implementing the broadband MFP methods aimed at localizing leaks.

Fig. 11(A) demonstrates the outcomes of applying MB-MFP(K) with $K = 1, 2, 3, 4$. It is observed that effective leak detection only occurs when $K \geq P = 2$. Especially, both leak locations and leak sizes can be estimated accurately, together with a correct identified number of leaks. In addition, Fig. 11(B) illustrates the results of employing MB-MFP(2) and MB-MFP'(2). MB-MFP'(2) demonstrates higher accuracy and effective side-lobe (or ambiguity) suppression, thereby confirming its superiority in practical leak detection scenarios within the laboratory environment. The Table 4 directly summarizes the results of additional two-leak detection cases. Similar to the results shown in Fig. 11, both MB-MFP methods achieve accurate estimations of leak locations and sizes (when $K \geq P$).

6. Conclusions

This study has presented a methodology using the modified broadband matched-field processing (MB-MFP) techniques to accurately and simultaneously estimate the number, sizes and locations of leaks in pressurized pipelines. The proposed techniques exhibit high efficiency and robustness against various noise types, including white, colored, and impulse noise. This robustness originates from the full utilization of the chosen-bandwidth signals and the normalization applied to the frequency-domain signals. Additionally, these techniques afford super-resolution capability (i.e., enabling the detection of leaks spaced closer than $0.5\lambda_{\min}$) and high-resolution estimation results. A key finding of our research is that when the number of leaks assumed in the model (K) equals or exceeds the actual number of leaks (P), the estimated leak locations reliably cluster around the real leak sites. This clustering effect facilitates not only the precise localization of leaks but also the accurate enumeration of them. The proposed methods are analyzed through

(a) 3D schematic:



(b) 2D Layout:

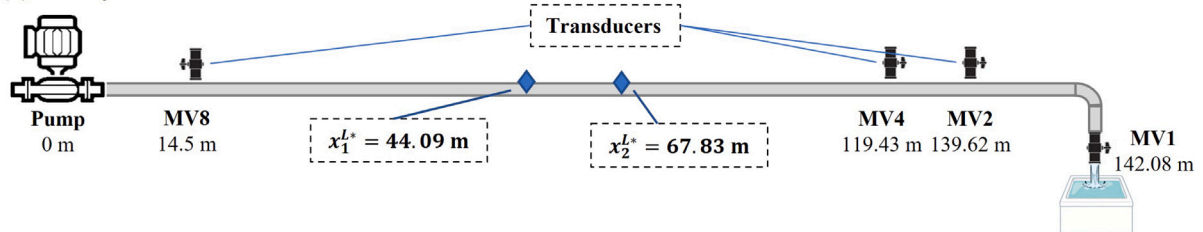


Fig. 9. The pipeline system in the Water Resources Research Laboratory of HKUST: (a) overall 3D schematic, and (b) simplified 2D layout.

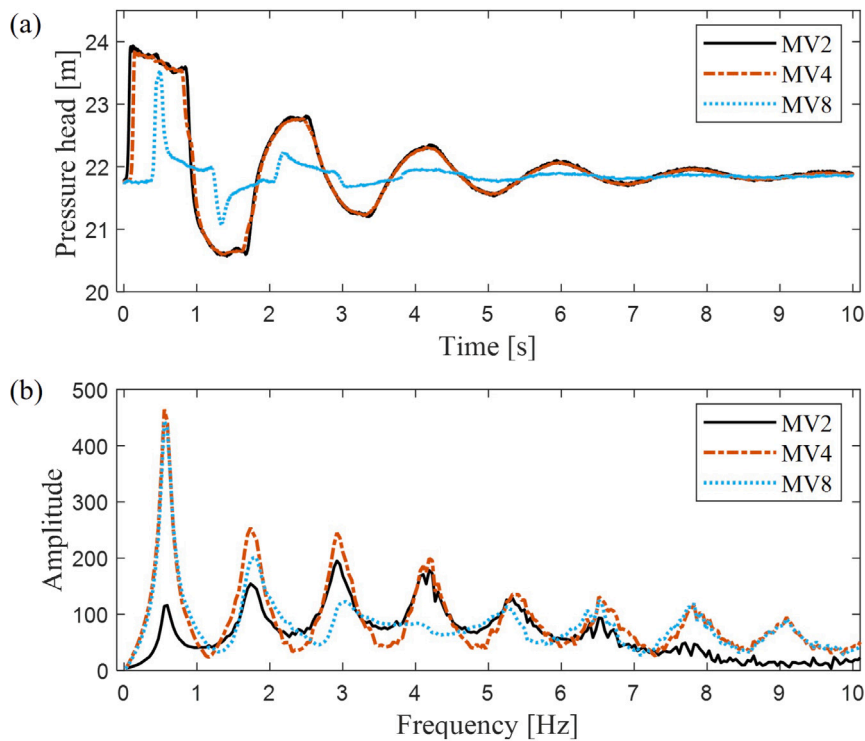


Fig. 10. (a) Measured time-domain pressure signals and (b) transformed frequency-domain signals at MV 2, MV 4, and MV 8.

Table 2

Key components of the HKUST experimental facility.

Component	Manufacturer/Model	Key specifications
Main pipe	Zhejiang Weixing New Bulding Materials Co., Ltd.	Material: HDPE, $L = 144$ m, $D = 79.2$ mm, $e = 5.4$ mm, Pressure rating: PE100, PN10, SDR = 17
Booster pump	KIKAWA, KRS16-3	Motor: Full-enclosed air-blast two-pole standed motor, Rated flow = $16 \text{ m}^3/\text{h}$, Stages = 3, Control: Variable frequency drive
Valves	Generic	Type: Manual Ball Valve, Orifice Diameter: 6.35 mm, Spec: Full-bore two way
Pressure transducers	UNIK, UNIK5000	Full-scale range of 0.5 MPa (5 bar), Accuracy = 0.04% (full scale best straight line)
Data acquisition	Dewesoft, IOLITE-6xSTG	Oversampling ratio (OSAR) = 24-bit OSAR with an anti-aliasing filter, Sampling rate = 20 kS/s
Ultrasonic flow meter	Dynasonics, DXNP-UHS-NN	Rate pulse: 0 to 1000 Hz, Total pulse duration: 33 ms, Accuracy: $\pm 1\%$ of reading or ± 0.01 FPS

Table 3

Viscoelastic coefficients and other parameters of the HDPE pipe system in the Water Resources Research Laboratory at HKUST.

Pipe length $L = 144$ m	Pipe internal diameter $D = 0.0792$ m
Elastic wave speed $a_0 = 365$ m/s	Darcy-Weisbach friction factor $f_D = 0.02$
Water density $\rho = 998.2 \text{ kg/m}^3$	Pipe wall thickness $e = 5.4 \times 10^{-3}$ m
Coefficients of the K-V model: $\tau_1 = 0.038$ s, $\tau_2 = 0.6$ s, $\tau_3 = 1.5$ s and $J_1 = 1.957 \times 10^{-10} \text{ Pa}^{-1}$, $J_2 = 1.0954 \times 10^{-10} \text{ Pa}^{-1}$, $J_3 = 9.05 \times 10^{-12} \text{ Pa}^{-1}$	

Table 4Predicted leak locations and sizes comparison. Here, P- n denotes the predicted result from the n -leak model.

Cases	Reference	Leak locations and sizes
Case 1	Actual	$\mathbf{x}^{L*} = (44.09, 67.83) \text{ m}$, $\mathbf{s}^{L*} = (1.75, 2.76) \times 10^{-5} \text{ m}^2$
	P-1	$\mathbf{x}^L = 73 \text{ m}$, $\mathbf{s}^L = 3.28 \times 10^{-5} \text{ m}^2$
	P-2	$\mathbf{x}^L = (45, 68) \text{ m}$, $\mathbf{s}^L = (1.67, 2.85) \times 10^{-5} \text{ m}^2$
	P-3	$\mathbf{x}^L = (42, 60, 68) \text{ m}$, $\mathbf{s}^L = (1.72, 0.01, 2.82) \times 10^{-5} \text{ m}^2$
	P-4	$\mathbf{x}^L = (44, 44, 68, 68) \text{ m}$, $\mathbf{s}^L = (0.56, 1.11, 0.83, 2.02) \times 10^{-5} \text{ m}^2$
Case 2	Actual	$\mathbf{x}^{L*} = (44.09, 98.86) \text{ m}$, $\mathbf{s}^{L*} = (2.12, 1.86) \times 10^{-5} \text{ m}^2$
	P-1	$\mathbf{x}^L = 65 \text{ m}$, $\mathbf{s}^L = 3.13 \times 10^{-5} \text{ m}^2$
	P-2	$\mathbf{x}^L = (44, 99) \text{ m}$, $\mathbf{s}^L = (2.01, 1.92) \times 10^{-5} \text{ m}^2$
	P-3	$\mathbf{x}^L = (11, 43, 98) \text{ m}$, $\mathbf{s}^L = (0.01, 2.08, 1.86) \times 10^{-5} \text{ m}^2$
	P-4	$\mathbf{x}^L = (44, 44, 99, 99) \text{ m}$, $\mathbf{s}^L = (1.02, 1.14, 0.71, 1.21) \times 10^{-5} \text{ m}^2$
Case 3	Actual	$\mathbf{x}^{L*} = (67.83, 98.86) \text{ m}$, $\mathbf{s}^{L*} = (2.45, 1.67) \times 10^{-5} \text{ m}^2$
	P-1	$\mathbf{x}^L = 49 \text{ m}$, $\mathbf{s}^L = 3.55 \times 10^{-5} \text{ m}^2$
	P-2	$\mathbf{x}^L = (68, 99) \text{ m}$, $\mathbf{s}^L = (2.40, 1.66) \times 10^{-5} \text{ m}^2$
	P-3	$\mathbf{x}^L = (68, 76, 98) \text{ m}$, $\mathbf{s}^L = (2.39, 0.01, 1.76) \times 10^{-5} \text{ m}^2$
	P-4	$\mathbf{x}^L = (68, 68, 99, 99) \text{ m}$, $\mathbf{s}^L = (0.62, 1.77, 0.89, 0.82) \times 10^{-5} \text{ m}^2$

comprehensive numerical simulations and validated with experimental data from laboratory settings.

Future work is planned to focus on applying the MB-MFP techniques for leak detection in real-field experiments and handling the relevant uncertainties. We are currently conducting fieldwork for 6 discrete District Metering Areas (DMAs) of various scales to validate the methodology. The uncertainties include insufficient knowledge of the system's structure and device behavior, and imprecise measurement of wave speed, friction factor, and steady-state discharge, which may affect the accuracy of the leak localization. Moreover, the fundamental Matched-Field Processing (MFP) framework is inherently generalizable to more complex network topologies, such as one with branches, loops, or variable diameters. One would simply replace the single-pipe transfer matrix model used here with a more sophisticated network hydraulic model. The core MB-MFP and MB-MFP' algorithms for matching, localization, and side-lobe suppression would remain the same. Therefore, the extension of our proposed MB-MFP techniques to complex networks is also a promising and logical direction for future research.

CRedit authorship contribution statement

Can Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data

curation, Conceptualization. **Muhammad Waqar:** Writing – review & editing, Supervision, Data curation. **Moez Louati:** Writing – review & editing, Supervision, Funding acquisition. **Mohamed Salah Ghidaoui:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

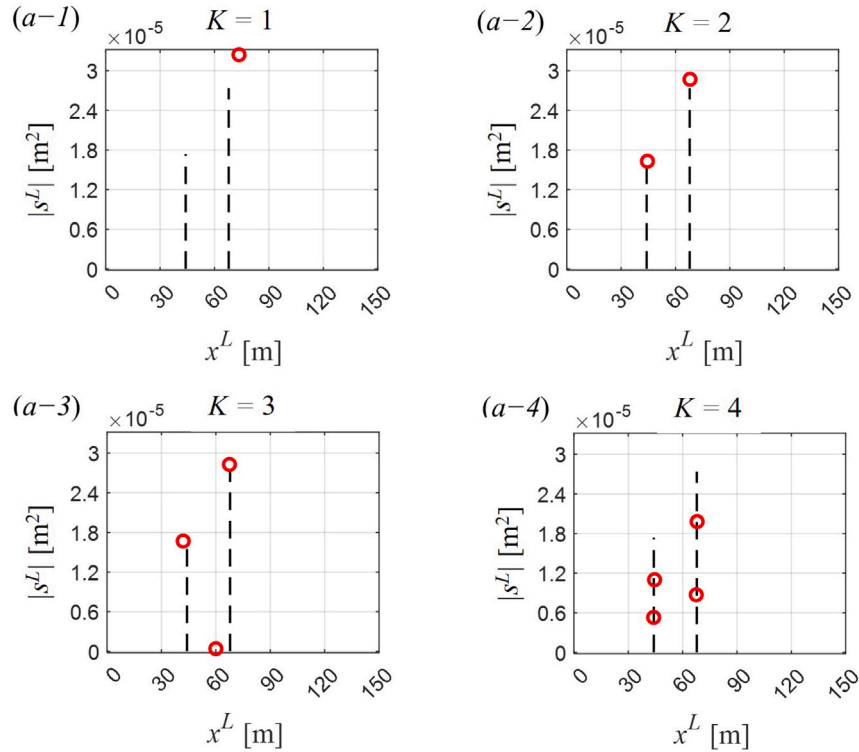
Acknowledgments

This work has been supported by research grants from the Research Grant Council of Hong Kong SAR (Project No.: 16213823 and 16214219), Chinese Estates Professorship in Engineering (No.: R8031), the Innovation and Technology Commission, Hong Kong (Projects No.: ITS/033/22MS & ITS/157/22), and the HKUST Postgraduate Studentship.

Data availability

Data will be made available on request.

(A) Results of leak localization MF-MFP for $P = 2$ and $\mathbf{x}^{L*} = [44.09 \text{ m}, 67.83 \text{ m}]$:



(B) Results of leak localization for $K = P = 2$ and $\mathbf{x}^{L*} = [44.09 \text{ m}, 67.83 \text{ m}]$

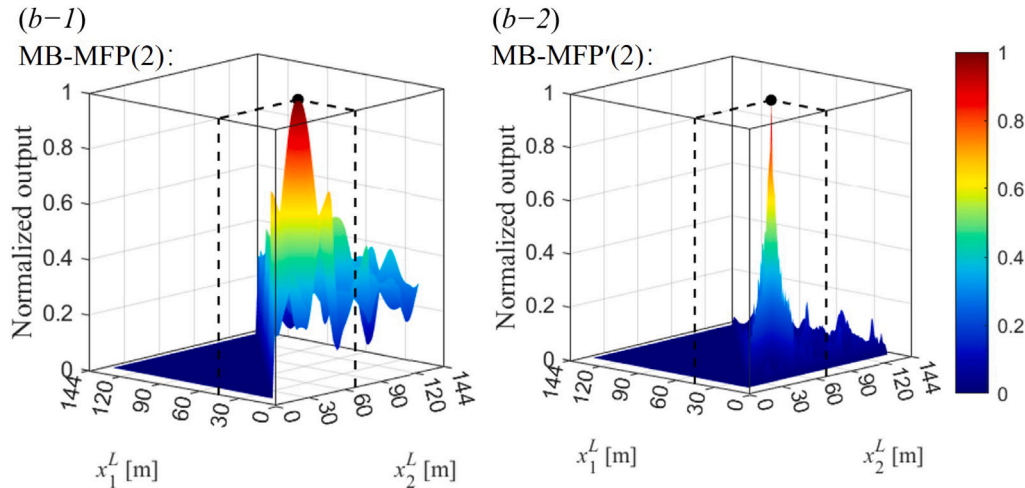


Fig. 11. Leak localization results of using MB-MFP(K) and MB-MFP'(K) with (a-1) $K = 1$, (a-2) $K = 2$, (a-3) $K = 3$, and (a-4) $K = 4$. The real leak locations are $\mathbf{x}^{L*} = [44.09 \text{ m}, 67.83 \text{ m}]$ and represented by the black vertical dashed lines. The triangles represent the estimated leak locations. In addition, the results shown in (B) present the capability of MB-MFP(2) and MB-MFP'(2) in suppressing the side-lobes.

References

- Brunone, B.B., 1999. Transient test-based technique for leak detection in outfall pipes. *J Water Resour Plan Manag.* 125, 302.
- Brunone, B., Cipolla, S., Greco, R., Meniconi, S., Vurro, M., 2008. Leak detection in water distribution networks using hydraulic modelling and transient analysis. *Urban Water J.* 5, 321.
- Brunone, B., Maietta, F., Capponi, C., Keramat, A., Meniconi, S., 2022. A review of physical experiments for leak detection in water pipes through transient tests for addressing future research. *J. Hydraul. Res.* 60, 894.
- Chaudhry, M.H., 2014. *Applied Hydraulic Transients*. Springer, New York, NY.
- Che, T.C., Duan, H.F., Lee, P.J., 2021. Transient wave-based methods for anomaly detection in fluid pipes: A review. *Mech. Syst. Signal Process.* 160, 107874.
- Che, T.C., Wang, X., Ghidaoui, M.S., 2022a. Leak localization in looped pipe networks based on a factorized transient wave model: Theoretical framework. *Water Resour. Res.* 58, e2021WR031364.
- Che, T.C., Wang, X., Louati, M., Zhou, L., Ghidaoui, M.S., 2022b. Leak localization in looped pipe networks based on a factorized transient wave model: 2 regularization of ill-conditioned problems. *Water Res. Res.* 58, e2022WR033293.
- Cheng, M., Li, J., 2023. Optimal sensor placement for leak location in water distribution networks: A feature selection method combined with graph signal processing. *Water Res.* 242, 120313.
- Dubey, A., Li, Z., Lee, P., Murch, R., 2019. Measurement and characterization of acoustic noise in water pipeline channels. *IEEE Access* 7, 56890.
- Ferrante, M., Brunone, B., Asce, M., Meniconi, S., 2007. Wavelets for the analysis of transient pressure signals for leak detection. *J. Hydraul. Eng.* 133, 1274.
- Ferrante, M., Brunone, B., Jim Yeh, T.C., 2002. Uncertainty analysis of transient unsaturated flow in bounded domain. *Water Resour. Res.* 38, 6.

- Fink, M., 2006. Time-reversal acoustics in complex environments. <http://dx.doi.org/10.1190/1.221535671>.
- Guibert, R., Bayle, A., Plouraboué, F., 2023. Geolocalization of water-waves origin within water distribution networks using time reversal of first event detection. *Water Res.* 230, 119538.
- Lin, J., Wang, X., Ghidaoui, M.S., 2021. Multi-sensor fusion for transient-based pipeline leak localization in the Dempster-Shafer evidence framework. *Water Resour. Res.* 57, e2021WR029926.
- Liu, R., Zayed, T., Xiao, R., 2024. Advanced acoustic leak detection in water distribution networks using integrated generative model. *Water Res.* 254, 121434.
- Livezey, J.A., Bujan, A.F., Sommer, F.T., 2016. Learning overcomplete, low coherence dictionaries with linear inference. *J. Mach. Learn. Res.* 20.
- Lo, M., Karuppanan, S., Ovinis, M., 2021. Failure pressure prediction of a corroded pipeline with longitudinally interacting corrosion defects subjected to combined loadings using FEM and ANN. *J. Mar. Sci. Eng.* 9, 281, 2021 9, Page 281.
- Louati, M., Ghidaoui, M.S., Tekitek, M.M., Lee, P.J., 2020. Wave-leak interaction in a simple pipe system. *J. Hydraul. Eng.* 146, 04020013.
- Meniconi, S., Capponi, C., Frisinghelli, M., Brunone, B., 2021. Leak detection in a real transmission main through transient tests: Deeds and misdeeds. *Water Resour. Res.* 57, e2020WR027838.
- Meniconi, S., Rubin, A., Tirello, L., Doro, A., Brunone, B., Capponi, C., 2024. Mapping pressure surge source in urban water networks: Integrating low- and high-frequency pressure data with an illustrative real case study. *Water Resour. Res.* 60, e2023WR036773.
- Nestleroth, B.J., Flamberg, S., Condit, W., Matthews, J., Wang, L., Chen, A.S.C., 2012. Field demonstration of innovative condition assessment technologies for water mains: Leak detection and location.
- OMS Ltd, Pipeline Defects – How to Avoid Them, <https://www.omsmeasure.com/blog/pipeline-defects-how-to-avoid-them>.
- Wang, X., 2021. Uncertainty quantification and global sensitivity analysis for transient wave propagation in pressurized pipes. *Water Resour. Res.* 57, e2020WR028975.
- Wang, X., Ghidaoui, M.S., 2018a. Identification of multiple leaks in pipeline: Linearized model, maximum likelihood, and super-resolution localization. *Mech. Syst. Signal Process.* 107, 529.
- Wang, X., Ghidaoui, M.S., 2018b. Pipeline leak detection using the matched-field processing method. *J. Hydraul. Eng.* 144, 04018030.
- Wang, X., Ghidaoui, M.S., 2019. Identification of multiple leaks in pipeline II: Iterative beamforming and leak number estimation. *Mech. Syst. Signal Process.* 119, 346.
- Wang, X., Ghidaoui, M.S., Lin, J., 2019a. Identification of multiple leaks in pipeline III: Experimental results. *Mech. Syst. Signal Process.* 130, 395.
- Wang, X., Palomar, D.P., Zhao, Licheng, Ghidaoui, M.S., Asce, M., Murch, R.D., 2019b. Spectral-based methods for pipeline leakage localization. *J. Hydraul. Eng.* 145, 04018089.
- Waqar, M., Louati, M., Ghidaoui, M.S., 2022. Time-reversal of water-hammer waves. *J. Hydraul. Res.* 60, 25.
- Waqar, M., Louati, M., Ghidaoui, M.S., 2023. Time-reversal technique for pipeline defect detection. *Water Res.* 243, 120375.
- Waqar, M., Louati, M., Wang, Xun, Ghidaoui, M.S., Asce, M., 2021. Model-free matched field processing for condition assessment of pressurized pipes. *J. Water Res. Plan Manag.* 147, 04021066.
- Wu, Y., Zheng, F., Yang, Y., Zhang, K., Du, K., Duan, H., Savic, D., Kapelan, Z., 2025. Enhancing inflow and infiltration detection in urban sewer networks with a new deterministic sensor placement method. *Water Res.* 268, 122591.
- Xu, C., Waqar, M., Louati, M., Ghidaoui, M.S., 2024. High-resolution pipeline leak localization using the side-lobes of the defect detection functional of matched-field processing. *Mech. Syst. Signal Process.* 217, 111529.
- Zhang, H., Sun, M., Zhang, J., Zhang, Y., Li, B., Zhai, K., 2023. Study on assessment method of failure pressure for pipelines with colony corrosion defects based on failure location. *Processes* 11, 3134, 2023 11, Page 3134.